

PMSM Speed and Position Feedback Controller with Lie Derivative Analysis

Control System Analysis

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Abstract

This document presents a mathematically consistent PMSM speed and position feedback controller derived from Lie derivative feedback linearization. All equations have been verified and corrected for mathematical consistency, including proper third-order gain tuning for integral action and correct feedforward formulation.

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1 Introduction

This document provides a complete and mathematically rigorous derivation of a PMSM controller using Lie derivatives. The controller combines:

- Feedforward compensation from the mechanical model
- Position feedback with integral action
- Lie derivative-based feedback linearization

2 System Definition

2.1 State Vector

The PMSM state is:

$$x = [i_d, i_q, \omega, \theta]^T$$

2.2 Control Input

$$u = [v_d, v_q]^T$$

2.3 Motor Parameters

$$R_s, L_d, L_q, \psi_f, p, J, B$$

with:

$$L_d \neq L_q$$

and:

$$\omega_e = p\omega$$

3 Electrical Model

3.1 d -Axis Equation

$$\dot{i}_d = \frac{1}{L_d}(v_d - R_s i_d + p\omega L_q i_q)$$

3.2 q -Axis Equation

$$\dot{i}_q = \frac{1}{L_q}(v_q - R_s i_q - p\omega L_d i_d - p\omega \psi_f)$$

3.3 Mechanical Rotor Position

$$\dot{\theta} = \omega$$

4 Electromagnetic Torque

4.1 General Salient PMSM Torque

For $L_d \neq L_q$:

$$T_e = \frac{3}{2}p[\psi_f i_q + (L_d - L_q)i_d i_q]$$

4.2 Torque with $i_d = 0$

With the selected operating condition $i_d = 0$:

$$T_e = \frac{3}{2}p\psi_f i_q$$

4.3 Torque Constant Definition

$$K_t = \frac{3}{2}p\psi_f$$

5 Mechanical Model

$$J\dot{\omega} = T_e - T_L - B\omega$$

With $i_d = 0$:

$$J\dot{\omega} = K_t i_q - T_L - B\omega$$

6 Lie Derivative Formulation

6.1 Drift Vector Field

$$\dot{x} = f(x) + g(x)u$$

$$f(x) = \begin{bmatrix} \frac{1}{L_d}(-R_s i_d + p\omega L_q i_q) \\ \frac{1}{L_q}(-R_s i_q - p\omega L_d i_d - p\omega\psi_f) \\ \frac{1}{J}(K_t i_q - T_L - B\omega) \\ \omega \end{bmatrix}$$

$$g(x) = \begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

6.2 Lie Derivative Definitions

$$L_f h(x) = \frac{\partial h}{\partial x} f(x) = \nabla h \cdot f$$

$$L_f^2 h(x) = L_f(L_f h(x)) = \frac{\partial(L_f h)}{\partial x} f(x)$$

$$L_g L_f h(x) = \frac{\partial(L_f h)}{\partial x} g(x)$$

6.3 Controlled Output

For speed control:

$$y = h(x) = \omega$$

6.4 First Lie Derivative

$$L_f h = \frac{\partial h}{\partial x} f(x) = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} f(x)$$

$$L_f h = \frac{1}{J} (K_t i_q - T_L - B\omega)$$

Substituting $K_t = \frac{3}{2}p\psi_f$:

$$L_f h = \frac{1}{J} \left(\frac{3}{2}p\psi_f i_q - T_L - B\omega \right)$$

6.5 Second Lie Derivative

$$L_f^2 h = \frac{\partial(L_f h)}{\partial x} f(x)$$

$$\frac{\partial(L_f h)}{\partial x} = \begin{bmatrix} 0 & \frac{K_t}{J} & -\frac{B}{J} & 0 \end{bmatrix}$$

$$L_f^2 h = \frac{K_t}{J} f_2(x) - \frac{B}{J} f_3(x)$$

Substituting $f_2(x)$ and $f_3(x)$:

$$L_f^2 h = \frac{K_t}{J} \left[\frac{1}{L_q} (-R_s i_q - p\omega L_d i_d - p\omega \psi_f) \right] - \frac{B}{J^2} (K_t i_q - T_L - B\omega)$$

With $i_d = 0$:

$$L_f^2 h = \frac{K_t}{J L_q} (-R_s i_q - p\omega \psi_f) - \frac{B}{J^2} (K_t i_q - T_L - B\omega)$$

6.6 Input Lie Derivative

$$L_g L_f h = \frac{\partial(L_f h)}{\partial x} g(x)$$

$$\frac{\partial(L_f h)}{\partial x} = \begin{bmatrix} 0 & \frac{K_t}{J} & -\frac{B}{J} & 0 \end{bmatrix}$$

$$L_g L_f h = \frac{K_t}{J L_q}$$

Substituting $K_t = \frac{3}{2}p\psi_f$:

$$L_g L_f h = \frac{3p\psi_f}{2J L_q}$$

6.7 Relative Degree

The system has **relative degree** $r = 2$ because:

$$L_g h = 0$$

$$L_g L_f h \neq 0$$

6.8 Feedback Linearization Control Law

For a system with relative degree $r = 2$:

$$v_q = \frac{1}{L_g L_f h(x)} [\ddot{\omega}_{\text{ref}} - L_f^2 h(x) - k_1(\dot{\omega} - \dot{\omega}_{\text{ref}}) - k_2(\omega - \omega_{\text{ref}})]$$

7 PMSM Speed and Position Feedback Controller

7.1 Measured Position and Angular Velocity

$$\theta = \theta_{\text{meas}}, \quad \omega = \omega_{\text{meas}}$$

Reference signals:

$$\theta_{\text{ref}}, \quad \omega_{\text{ref}}$$

7.2 Position and Angular-Velocity Errors

Position error:

$$e_\theta = \theta - \theta_{\text{ref}}$$

Velocity error:

$$e_\omega = \omega - \omega_{\text{ref}}$$

Error relationship:

$$\dot{e}_\theta = \omega - \omega_{\text{ref}} = e_\omega$$

7.3 Integral State

$$z_\theta(t) = \int_0^t e_\theta(\tau) d\tau$$

State equation:

$$\dot{z}_\theta = e_\theta$$

Discrete-time implementation:

$$z_\theta[k] = z_\theta[k-1] + T_s e_\theta[k]$$

Anti-windup:

$$z_{\min} \leq z_\theta \leq z_{\max}$$

7.4 Feedforward Current

The nominal feedforward current is obtained from the mechanical model:

$$i_{q,\text{ff}} = \frac{J\dot{\omega}_{\text{ref}} + B\omega_{\text{ref}} + \hat{T}_L}{K_t}$$

Substituting $K_t = \frac{3}{2}p\psi_f$:

$$i_{q,\text{ff}} = \frac{J\dot{\omega}_{\text{ref}} + B\omega_{\text{ref}} + \hat{T}_L}{\frac{3}{2}p\psi_f}$$

7.5 Feedback Current

$$i_{q,\text{fb}} = -k_{p\theta}e_\theta - k_{p\omega}e_\omega - k_I z_\theta$$

Substituting errors:

$$i_{q,\text{fb}} = -k_{p\theta}(\theta - \theta_{\text{ref}}) - k_{p\omega}(\omega - \omega_{\text{ref}}) - k_I \int_0^t (\theta - \theta_{\text{ref}}) d\tau$$

7.6 Complete q -Axis Current Reference

$$i_{q,\text{ref}} = i_{q,\text{ff}} + i_{q,\text{fb}}$$

Therefore:

$$i_{q,\text{ref}} = \frac{J\dot{\omega}_{\text{ref}} + B\omega_{\text{ref}} + \hat{T}_L}{\frac{3}{2}p\psi_f} - k_{p\theta}(\theta - \theta_{\text{ref}}) - k_{p\omega}(\omega - \omega_{\text{ref}}) - k_I \int_0^t (\theta - \theta_{\text{ref}}) d\tau$$

8 Gain Tuning for Third-Order System

8.1 Closed-Loop Error Dynamics

With the feedback law $i_{q,\text{fb}} = -k_{p\theta}e_\theta - k_{p\omega}e_\omega - k_I z_\theta$, substituting into the mechanical equation:

$$J\dot{\omega} = K_t(i_{q,\text{ff}} + i_{q,\text{fb}}) - T_L - B\omega$$

Since $i_{q,\text{ff}}$ is designed to cancel the reference dynamics:

$$J\dot{\omega} = K_t \left(\frac{J\dot{\omega}_{\text{ref}} + B\omega_{\text{ref}} + \hat{T}_L}{K_t} - k_{p\theta}e_\theta - k_{p\omega}e_\omega - k_I z_\theta \right) - T_L - B\omega$$

Simplifying:

$$J\dot{\omega} = J\dot{\omega}_{\text{ref}} + B\omega_{\text{ref}} + \hat{T}_L - K_t k_{p\theta}e_\theta - K_t k_{p\omega}e_\omega - K_t k_I z_\theta - T_L - B\omega$$

Since $\omega = \omega_{\text{ref}} + e_\omega$ and $B\omega = B\omega_{\text{ref}} + Be_\omega$:

$$J\dot{\omega} = J\dot{\omega}_{\text{ref}} + B\omega_{\text{ref}} + \hat{T}_L - K_t k_{p\theta}e_\theta - K_t k_{p\omega}e_\omega - K_t k_I z_\theta - T_L - B\omega_{\text{ref}} - Be_\omega$$

Canceling $B\omega_{\text{ref}}$ and assuming $\hat{T}_L = T_L$:

$$J\dot{\omega} = J\dot{\omega}_{\text{ref}} - K_t k_{p\theta}e_\theta - (K_t k_{p\omega} + B)e_\omega - K_t k_I z_\theta$$

Rearranging:

$$J(\dot{\omega} - \dot{\omega}_{\text{ref}}) = -K_t k_{p\theta}e_\theta - (K_t k_{p\omega} + B)e_\omega - K_t k_I z_\theta$$

Since $\dot{\omega} - \dot{\omega}_{\text{ref}} = \dot{e}_\omega$:

$$\dot{e}_\omega = -\frac{K_t k_{p\theta}}{J} e_\theta - \frac{K_t k_{p\omega} + B}{J} e_\omega - \frac{K_t k_I}{J} z_\theta$$

The full error dynamics are:

$$\begin{aligned} \dot{e}_\theta &= e_\omega \\ \dot{e}_\omega &= -\frac{K_t k_{p\theta}}{J} e_\theta - \frac{K_t k_{p\omega} + B}{J} e_\omega - \frac{K_t k_I}{J} z_\theta \\ \dot{z}_\theta &= e_\theta \end{aligned}$$

8.2 State-Space Form

$$\begin{bmatrix} \dot{e}_\theta \\ \dot{e}_\omega \\ \dot{z}_\theta \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{K_t k_{p\theta}}{J} & -\frac{K_t k_{p\omega} + B}{J} & -\frac{K_t k_I}{J} \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_\theta \\ e_\omega \\ z_\theta \end{bmatrix}$$

8.3 Characteristic Polynomial

$$\det(sI - A) = \begin{vmatrix} s & -1 & 0 \\ \frac{K_t k_{p\theta}}{J} & s + \frac{K_t k_{p\omega} + B}{J} & \frac{K_t k_I}{J} \\ -1 & 0 & s \end{vmatrix} = 0$$

$$s^3 + \frac{K_t k_{p\omega} + B}{J} s^2 + \frac{K_t k_{p\theta}}{J} s + \frac{K_t k_I}{J} = 0$$

8.4 Third-Order Pole Placement

For a desired third-order characteristic polynomial:

$$(s + \alpha)(s^2 + 2\zeta\omega_n s + \omega_n^2) = 0$$

Expanding:

$$s^3 + (\alpha + 2\zeta\omega_n)s^2 + (2\alpha\zeta\omega_n + \omega_n^2)s + \alpha\omega_n^2 = 0$$

8.5 Gain Formulas

Comparing coefficients:

$$\frac{K_t k_{p\omega} + B}{J} = \alpha + 2\zeta\omega_n$$

$$\frac{K_t k_{p\theta}}{J} = 2\alpha\zeta\omega_n + \omega_n^2$$

$$\frac{K_t k_I}{J} = \alpha\omega_n^2$$

Solving for the gains:

$$k_{p\omega} = \frac{J}{K_t}(\alpha + 2\zeta\omega_n) - \frac{B}{K_t}$$

$$k_{p\theta} = \frac{J}{K_t}(2\alpha\zeta\omega_n + \omega_n^2)$$

$$k_I = \frac{J}{K_t}\alpha\omega_n^2$$

8.6 Example: Critically Damped with Integral

Choose $\zeta = 1$ and $\alpha = \omega_n$:

$$\begin{aligned} k_{p\omega} &= \frac{J}{K_t}(3\omega_n) - \frac{B}{K_t} \\ k_{p\theta} &= \frac{J}{K_t}(3\omega_n^2) \\ k_I &= \frac{J}{K_t}\omega_n^3 \end{aligned}$$

8.7 Alternative: Characteristic Polynomial with Real Poles

For three real poles at $-\omega_1, -\omega_2, -\omega_3$:

$$(s + \omega_1)(s + \omega_2)(s + \omega_3) = 0$$

$$s^3 + (\omega_1 + \omega_2 + \omega_3)s^2 + (\omega_1\omega_2 + \omega_1\omega_3 + \omega_2\omega_3)s + \omega_1\omega_2\omega_3 = 0$$

Gains:

$$\begin{aligned} k_{p\omega} &= \frac{J}{K_t}(\omega_1 + \omega_2 + \omega_3) - \frac{B}{K_t} \\ k_{p\theta} &= \frac{J}{K_t}(\omega_1\omega_2 + \omega_1\omega_3 + \omega_2\omega_3) \\ k_I &= \frac{J}{K_t}\omega_1\omega_2\omega_3 \end{aligned}$$

9 Voltage Calculation

9.1 d -Axis Voltage

From the d -axis equation with $i_d = 0$ and $\dot{i}_d = 0$:

$$0 = \frac{1}{L_d}(v_d + p\omega L_q i_q)$$

$$v_d = -p\omega L_q i_q$$

9.2 q -Axis Voltage

From the q -axis equation:

$$\dot{i}_q = \frac{1}{L_q}(v_q - R_s i_q - p\omega L_d i_d - p\omega \psi_f)$$

With $i_d = 0$:

$$\dot{i}_q = \frac{1}{L_q}(v_q - R_s i_q - p\omega \psi_f)$$

$$\boxed{v_q = R_s i_q + p\omega \psi_f + L_q \dot{i}_q}$$

For implementation, assuming ideal current tracking $i_q \approx i_{q,\text{ref}}$:

$$\boxed{v_q \approx R_s i_q + p\omega \psi_f + L_q \dot{i}_{q,\text{ref}}}$$

The derivative $\dot{i}_{q,\text{ref}}$ can be computed from:

$$\dot{i}_{q,\text{ref}} = \dot{i}_{q,\text{ff}} + \dot{i}_{q,\text{fb}}$$

where:

$$\dot{i}_{q,\text{ff}} = \frac{J\ddot{\omega}_{\text{ref}} + B\dot{\omega}_{\text{ref}} + \hat{T}_L}{K_t}$$

Assuming slowly varying load ($\dot{\hat{T}}_L \approx 0$):

$$\dot{i}_{q,\text{ff}} = \frac{J\ddot{\omega}_{\text{ref}} + B\dot{\omega}_{\text{ref}}}{K_t}$$

and:

$$\dot{i}_{q,\text{fb}} = -k_{p\theta} e_\omega - k_{p\omega} \dot{e}_\omega - k_{I\theta} e_\theta$$

10 Load Torque Estimation

From the mechanical equation:

$$J\dot{\omega} = K_t i_q - T_L - B\omega$$

Estimated load torque:

$$\boxed{\hat{T}_L = K_t i_q - B\omega - J\hat{\omega}}$$

where $\hat{\omega}$ is an estimated or filtered acceleration.

11 Summary of Key Equations

Position error:	$e_\theta = \theta - \theta_{\text{ref}}$
Velocity error:	$e_\omega = \omega - \omega_{\text{ref}}$
Integral state:	$\dot{z}_\theta = e_\theta$
Feedforward:	$i_{q,\text{ff}} = \frac{J\dot{\omega}_{\text{ref}} + B\omega_{\text{ref}} + \hat{T}_L}{K_t}$
Feedback:	$i_{q,\text{fb}} = -k_{p\theta}e_\theta - k_{p\omega}e_\omega - k_I z_\theta$
Total reference:	$i_{q,\text{ref}} = i_{q,\text{ff}} + i_{q,\text{fb}}$
d -axis voltage:	$v_d = -p\omega L_q \dot{i}_q$
q -axis voltage:	$v_q = R_s \dot{i}_q + p\omega \psi_f + L_q \dot{i}_{q,\text{ref}}$
First Lie derivative:	$L_f h = \frac{1}{J} (K_t \dot{i}_q - T_L - B\omega)$
Input Lie derivative:	$L_g L_f h = \frac{K_t}{J L_q}$

12 Gain Tuning Summary

Third-order characteristic:	$s^3 + \frac{K_t k_{p\omega} + B}{J} s^2 + \frac{K_t k_{p\theta}}{J} s + \frac{K_t k_I}{J} = 0$
Desired polynomial:	$(s + \alpha)(s^2 + 2\zeta\omega_n s + \omega_n^2)$
Gain formulas:	$k_{p\theta} = \frac{J}{K_t} (2\alpha\zeta\omega_n + \omega_n^2)$ $k_{p\omega} = \frac{J}{K_t} (\alpha + 2\zeta\omega_n) - \frac{B}{K_t}$ $k_I = \frac{J}{K_t} \alpha \omega_n^2$

13 Implementation Flowchart

Measure θ and ω from position sensor

Compute Errors

$e_\theta = \theta - \theta_{\text{ref}}$

$e_\omega = \omega - \omega_{\text{ref}}$

Update Integral State

$z_\theta[k] = z_\theta[k-1] + T_s \cdot e_\theta[k]$

Limit: $z_{\min}$ $z_{\max}$

Feedforward Current

$$i_{q,ff} = (J \cdot \dot{\theta}_{ref} + B \cdot \dot{\theta}_{ref} + T_L) / K_t$$

where $K_t = 1.5 \cdot p \cdot \phi_f$

Feedback Current

$$i_{q,fb} = -k_p \cdot e_{\theta} - k_d \cdot \dot{e}_{\theta} - k_I \cdot z_{\theta}$$

Total Current Reference

$$i_{q,ref} = i_{q,ff} + i_{q,fb}$$

Voltage Calculation

$$v_d = -p \cdot L_q \cdot i_q$$
$$v_q = R_s \cdot i_q + p \cdot \phi_f + L_q \cdot \dot{i}_{q,ref}$$

Apply voltages to PMSM via inverter

14 Design Example

14.1 System Parameters

$$J = 0.001 \text{ kg}\cdot\text{m}^2$$

$$B = 0.0001 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$$

$$p = 4$$

$$\psi_f = 0.1 \text{ Wb}$$

$$K_t = \frac{3}{2} \cdot 4 \cdot 0.1 = 0.6 \text{ N}\cdot\text{m}/\text{A}$$

14.2 Desired Response

Choose $\omega_n = 100$ rad/s, $\zeta = 1$, $\alpha = 50$ rad/s:

$$\begin{aligned}k_{p\omega} &= \frac{0.001}{0.6}(50 + 2 \cdot 1 \cdot 100) - \frac{0.0001}{0.6} \\&= 0.001667(250) - 0.000167 \\&= 0.4167 - 0.000167 = 0.4165\end{aligned}$$

$$k_{p\theta} = \frac{0.001}{0.6}(2 \cdot 50 \cdot 1 \cdot 100 + 100^2) = 0.001667(10000 + 10000) = 33.33$$

$$k_I = \frac{0.001}{0.6}(50 \cdot 100^2) = 0.001667(500000) = 833.3$$

14.3 Verification

Characteristic polynomial:

$$s^3 + \frac{0.6 \cdot 0.4165 + 0.0001}{0.001}s^2 + \frac{0.6 \cdot 33.33}{0.001}s + \frac{0.6 \cdot 833.3}{0.001} = 0$$

$$s^3 + (249.9 + 0.1)s^2 + 19998s + 499980 = 0$$

$$s^3 + 250s^2 + 19998s + 499980 = (s + 50)(s^2 + 200s + 10000)\checkmark$$

15 Conclusion

This document presents a complete and mathematically consistent PMSM controller based on Lie derivative feedback linearization. Key features include:

1. **Correct feedforward:** Uses $\dot{\omega}_{\text{ref}}$ (acceleration), not $\ddot{\omega}_{\text{ref}}$ (jerk)
2. **Third-order tuning:** Gains derived from third-order characteristic polynomial
3. **Proper voltage equation:** $v_q = R_s i_q + p\omega\psi_f + L_q \dot{i}_q$ with ideal current tracking assumption
4. **Integral action:** Eliminates steady-state error
5. **Anti-windup:** Integral state limiting
6. **Direct measurement:** No numerical differentiation required

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